

Bridging Skilled Migration & Australian ICT Hiring Via Deterministic Continuous Mathematics

Eliminating the overseas "resume vocabulary penalty" for Australian visa holders through automated ANZSCO taxonomy harmonization, zero-PII merit scoring, and explainable algebraic ranking.

1 The Australian Skilled Migration Problem

MARKET GAP

42%

AUSTRALIAN ICT TALENT SHORTAGE

68%

MIGRANT UNDEREMPLOYMENT RATE

Zero

DEMOGRAPHIC PII LEAKS

The Core Failure: Australian employers rely on rigid keyword-filtering ATS algorithms that reject highly qualified overseas professionals (subclasses 189, 190, 491) due to non-standard titles (e.g. "MIS Executive" vs "Data Analyst").

The Jinder Remedy: Autonomous translation to the Australian Bureau of Statistics (ABS ANZSCO 2026) taxonomy graph coupled with deterministic mathematical match engines operating under ASD-STE100 (Simplified Technical English) specification.

2 Medallion 3-Tier Data Pipeline

DATA ARCHITECTURE

BRONZE TIER

Raw Ingestion

- Raw CV text & JD payloads
- Strict size caps (10MB / file)
- Immutable audit event log
- PST / AEST normalized timestamps

SILVER TIER

Harmonized Graph

- ANZSCO 2026 ontology normalization
- 172 verified technical competencies
- 53 occupations & 20 specialisations
- Evidence anchoring & confidence

GOLD TIER

Zero-PII Match Graph

- SQLite 3 WAL (22 relational tables)
- Gaussian Z-scores & Jaccard index
- Statutory blocker gatekeeping
- Instant recruiter shortlist delivery

3 The 6 Canonical Algebraic Formulations (Intelligence Engine V2)

ASD-STE100 CERTIFIED • <5MS SLA DETERMINISM

F-01 • SMF

0.82 ms

Skill Match Frequency & Coverage

$$SMF = \frac{\sum w_i \min\left(1.0, \frac{L_{c,i}}{L_{j,i}}\right)}{\sum w_k \times 100}$$

Weights mandatory (w=2.0) vs preferred skills (w=1.0) with verified proficiency depth.

F-02 • GSI & JRS

0.45 ms

Gap Severity Index & Net Readiness

$$GSI = \sum \beta_g \cdot \text{Sev}(g), \quad JRS = \max(0, SMF - GSI)$$

Penalises statutory blockers (β=15.0) heavily while forgiving learnable tools (β=5.0).

F-03 • JPI

0.65 ms

Job Proximity Index Mobility

$$JPI = 100 \left[0.70 \frac{|S_A \cap S_B|}{|S_A \cup S_B|} + 0.30 \delta_{\text{dom}} \right]$$

Measures cross-career mobility using Jaccard vector similarity plus domain taxonomy affinity.

F-04 • RMS

1.10 ms

Relative Merit Score (Zero-PII Pool)

$$Z_c = \frac{SMF_c - \mu}{\sigma + \epsilon}, \quad RMS = \frac{1}{1 + e^{-1.702 Z_c}}$$

Gaussian Z-score standardization against active applicant pool with zero demographic bias.

F-05 • FRS

1.82 ms

Feed Ranking Score (Talent Feed)

$$FRS = 0.45 SMF + 0.25 JRS + 0.20 \text{Exp} + 0.10 \text{Mode}$$

Multi-objective ranking for candidate feed balancing skill fit, readiness, and work modes.

F-06 • TSS

1.35 ms

Employer Talent Search Score

$$TSS = 0.40 SMF + 0.25 \text{Sen} + 0.20 \text{Cred} + 0.15 \text{Rec}$$

Ranks candidate pool for hiring employers factoring capability match, seniority, and recency.

4 Dual-Role Portal Experience

PRODUCT ARCHITECTURE

TALENT FLOW • CANDIDATE VIEW

Interactive Radar & Match Transparency

- Dynamic 5-Axis Radar Chart:** Real-time SVG rendering of candidate capability vs Australian requisition benchmarks.
- Cross-Border Translation:** Displays overseas job mappings with explicit evidence citations and confidence levels.
- Net Wage Upside:** Continuous logistic salary progression index based on verified capability surplus.

EMPLOYER FLOW • RECRUITER VIEW

Anonymized Wildlife Talent Cards

- Zero-PII Wildlife Cards:** Candidate profiles masked under *Silver*, *Teal*, *Koala*, *Heron*. Australian wildlife personas (e.g. *Wombat*, *Emu*).
- Multi-Candidate Compare Tray:** Side-by-side benchmarking of up to 4 shortlisted profiles simultaneously.
- Instant Match Telemetry:** Live breakdown of mandatory requirement coverage vs statutory blockers.

5 Engineering Rigor & Verification

QUALITY PROVENANCE

737 / 737

TESTS PASSING (24S)

0

EXTERNAL PIP PACKAGES

0

NPM BUILD DEPENDENCIES

100%

DETERMINISTIC MATCH

Deployment Baseline: Built entirely on the **Python Standard Library** (http.server, socketserver, sqlite3) with a zero-build vanilla ES6 frontend. Deployed in Sydney (ap-southeast-2), compliant with the **Australian Privacy Principles (APP 1-13)**.